Lecture 6

Characters

Objectives. *Lecture 5 counted the copies of an irreducible U in V by* $\dim \mathrm{Hom}_G(U,V)$. *What does a representation's trace remember? Everything, up to isomorphism. Averaging on* $\mathrm{Hom}(U,V)$ *gives* $\dim \mathrm{Hom}_G(U,V) = \langle \chi_U, \chi_V \rangle_G$, *for the* characters $\chi(g) = \operatorname{tr} \rho(g)$, *functions on the classes; so the irreducible characters are orthonormal. With the characters of duals, tensor products and quotients, we build the table of* S_4.

By the end of this lecture you will be able to:

- Compute characters, and decide from their inner products whether a representation is irreducible and how it decomposes.
- Prove that $\dim \mathrm{Hom}_G(U,V) = \langle \chi_U, \chi_V \rangle_G$ by averaging on $\mathrm{Hom}(U,V)$, and deduce that the irreducible characters are orthonormal and that a representation is determined up to isomorphism by its character.
- Compute the characters of V^*, $V \otimes W$, $\mathrm{Sym}^2(V)$, $\Lambda^2(V)$ and of a lift from a quotient, and prove that $|\chi(g)| \leq \chi(e)$, with equality exactly when $\rho(g)$ is a scalar.
- Build the character table of S_4 from the trivial, sign and permutation representations and a lift from $S_4/V_4 \cong S_3$, and prove that it lists every irreducible character.

6.1 Characters and class functions

Throughout, G is finite, and representations are complex and finite-dimensional.

Definition 6.1 (Characters). The *character* of a representation (ρ, V) is the function

$$\chi_V : G \to \mathbb{C}, \qquad \chi_V(g) = \operatorname{tr} \rho(g).$$

Its *degree* is $\chi_V(e) = \dim V$. It is *irreducible* if V is, and *linear* if $\dim V = 1$; then it is the homomorphism $\rho : G \to \mathbb{C}^\times$ itself.

The trivial representation has the constant character 1_G.

Proposition 6.2 (First properties). *Let V and W be representations of G, and $g, h \in G$.*

(i) $\chi_V(hgh^{-1}) = \chi_V(g)$.
(ii) $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.
(iii) If $V \cong W$, then $\chi_V = \chi_W$. On $V \oplus W$, with G acting on each summand, $\chi_{V \oplus W} = \chi_V + \chi_W$.

Proof. (i) $\rho(hgh^{-1}) = \rho(h)\rho(g)\rho(h)^{-1}$, and conjugate matrices have the same trace.
(ii) By Corollary 5.14(ii), $\rho(g)$ is diagonalisable, with eigenvalues $c_1, \ldots, c_d$ that are roots

of unity; so $|c_i| = 1$ and $c_i^{-1} = \overline{c_i}$. The eigenvalues of $\rho(g^{-1}) = \rho(g)^{-1}$ are the c_i^{-1}, so

$$\chi_V(g^{-1}) = \sum_i c_i^{-1} = \sum_i \overline{c_i} = \overline{\chi_V(g)}.$$

(iii) Isomorphic representations differ by a change of basis (§ 5.1), which keeps traces. In a basis of V followed by one of W, the matrix of g on $V \oplus W$ is block diagonal, with blocks $D_V(g)$ and $D_W(g)$. ∎

Definition 6.3 (Class functions)**.** A *class function* is a function $\varphi : G \to \mathbb{C}$ that is constant on each conjugacy class. The class functions form a vector space $\mathrm{CF}(G)$, with the inner product

$$\langle \varphi, \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\varphi(g)}\, \psi(g).$$

The indicator functions of the classes are a basis of $\mathrm{CF}(G)$, so $\dim \mathrm{CF}(G)$ is the number of classes. The form is antilinear in the first slot and linear in the second, and $\langle \varphi, \varphi \rangle_G > 0$ for $\varphi \neq 0$.[1] Conjugacy invariance makes characters class functions. Summing over classes,

$$\langle \varphi, \psi \rangle_G = \frac{1}{|G|} \sum_{\text{classes } [g]} |[g]|\, \overline{\varphi(g)}\, \psi(g).$$

Example 6.4 (The characters of S_3)**.** From Example 1.16 and § 5.5:

class of	e	$(1\,2)$	$(1\,2\,3)$
size	1	3	2
1_G	1	1	1
sgn	1	−1	1
χ_E	2	0	−1

Example 6.5 (Guard: traces, not eigenvalues)**.** A character sums eigenvalues; in degree above 1 it need not be a homomorphism. $D_E((1\,2\,3))$ has characteristic polynomial $t^2 + t + 1$, with roots ζ, ζ^2 for $\zeta = \exp(2\pi \mathrm{i}/3)$, so

$$\chi_E\big((1\,2\,3)\big) = \zeta + \zeta^2 = -1.$$

Its square $(1\,3\,2)$ has the eigenvalues ζ^2, ζ^4, and

$$\chi_E\big((1\,2\,3)^2\big) = -1, \qquad \chi_E\big((1\,2\,3)\big)^2 = 1.$$

§ 6.3 and Lecture 9's Frobenius–Schur indicator need $\chi(g^2)$.

6.2 Averaging on Hom and orthonormality

For representations U and V, G acts on $\mathrm{Hom}(U,V)$ by $g \cdot T = \rho_V(g)\, T\, \rho_U(g)^{-1}$, with invariants $\mathrm{Hom}(U,V)^G = \mathrm{Hom}_G(U,V)$ (§ 5.2).

Lemma 6.6 (The character of Hom)**.** *The character of* $\mathrm{Hom}(U,V)$ *is*

$$\chi_{\mathrm{Hom}(U,V)}(g) = \overline{\chi_U(g)}\, \chi_V(g).$$

[1] Serre and James–Liebeck put the conjugate on the second slot.

Proof. In bases of U and V, $\mathrm{Hom}(U,V)$ is the space of $n \times m$ matrices, $n = \dim V$, $m = \dim U$, with basis the matrix units E_{kl} (1 in row k, column l, 0 elsewhere).

A trace. Let A be $n \times n$ and B be $m \times m$. The E_{kl}-coordinate of $AE_{kl}B$ is its (k,l) entry,

$$(AE_{kl}B)_{kl} = A_{kk}B_{ll},$$

and summing over k and l, the map $T \mapsto ATB$ has trace $\operatorname{tr} A \operatorname{tr} B$.

The character. Take $A = D_V(g)$ and $B = D_U(g^{-1})$. By Proposition 6.2(ii),

$$\chi_{\mathrm{Hom}(U,V)}(g) = \chi_V(g)\,\chi_U(g^{-1}) = \overline{\chi_U(g)}\,\chi_V(g).$$ ∎

Theorem 6.7 (Characters count intertwiners). *For representations U and V of G,*

$$\dim \mathrm{Hom}_G(U,V) = \langle \chi_U, \chi_V \rangle_G .$$

Proof. By Proposition 5.7(iii) on $\mathrm{Hom}(U,V)$, and Lemma 6.6,

$$\begin{aligned} \dim \mathrm{Hom}_G(U,V) &= \dim \mathrm{Hom}(U,V)^G \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_{\mathrm{Hom}(U,V)}(g) \\ &= \frac{1}{|G|} \sum_{g \in G} \overline{\chi_U(g)}\,\chi_V(g) \\ &= \langle \chi_U, \chi_V \rangle_G . \end{aligned}$$ ∎

In particular, $\langle \chi_U, \chi_V \rangle_G$ is a non-negative integer.

Choose one representative of each isomorphism class of irreducible representations; write $\mathrm{Irr}(G)$ for this set.

Corollary 6.8 (Orthonormality). *Let $U, U' \in \mathrm{Irr}(G)$, and let V and W be representations of G.*

(i) $\langle \chi_U, \chi_{U'} \rangle_G$ *is* 1 *if* $U = U'$, *and* 0 *otherwise.*

(ii) $m_U(V) = \langle \chi_U, \chi_V \rangle_G$, *and* $\chi_V = \sum_U m_U(V)\,\chi_U$, *a finite sum over the U with* $m_U(V) \neq 0$.

(iii) $V \cong W$ *if and only if* $\chi_V = \chi_W$.

(iv) $\langle \chi_V, \chi_V \rangle_G = \sum_U m_U(V)^2$. *So V is irreducible if and only if* $\langle \chi_V, \chi_V \rangle_G = 1$.

(v) The irreducible characters are linearly independent in $\mathrm{CF}(G)$. *So* $\mathrm{Irr}(G)$ *has at most as many elements as G has classes; and if k irreducible representations have pairwise distinct characters, and k is the number of classes, they are all of* $\mathrm{Irr}(G)$, *up to isomorphism.*

Proof. (i) By Theorem 6.7 and Schur's lemma (Theorem 5.11(ii)), $\dim \mathrm{Hom}_G(U,U')$ is 1 or 0.

(ii) The first claim is Theorem 6.7 with Corollary 5.13. By Maschke's theorem, $V = U_1 \oplus \cdots \oplus U_k$ with each U_j irreducible, so $\chi_V = \sum_j \chi_{U_j}$ by Proposition 6.2(iii); and exactly $m_U(V)$ of the U_j are isomorphic to U.

(iii) One direction is Proposition 6.2(iii). Conversely, if $\chi_V = \chi_W$, the multiplicity formula gives the same multiplicities, and $V \cong W$ by Corollary 5.13.

(iv) Expand both characters using the multiplicity formula, then orthonormality:

$$\begin{aligned} \langle \chi_V, \chi_V \rangle_G &= \sum_{U,U'} m_U(V)\,m_{U'}(V)\,\langle \chi_U, \chi_{U'} \rangle_G \\ &= \sum_U m_U(V)^2. \end{aligned}$$

A sum of squares of non-negative integers is 1 exactly when one term is 1 and the others are 0, that is, when $V \cong U$ for one U (Corollary 5.13).

(v) *Independence.* If $\sum_U a_U \chi_U = 0$, a finite sum, take its inner product with $\chi_{U'}$: orthonormality gives $a_{U'} = 0$.

The count. So there are at most $\dim \mathrm{CF}(G)$ irreducible characters, hence at most that many irreducibles, since characters determine isomorphism classes.

Completeness. If $U_1, \dots, U_k$ have pairwise distinct characters, they have orthonormal characters. If $k = \dim \mathrm{CF}(G)$, these form a basis. Any further irreducible U has character orthogonal to the basis, hence zero. But $\chi_U(e) = \dim U \geq 1$. ∎

Example 6.9 (S_3 again)**.** From the table of Example 6.4,

$$\langle \chi_E, \chi_E \rangle_{S_3} = \frac{1}{6}(4 + 3 \cdot 0 + 2 \cdot 1) = 1,$$

so E is irreducible by the character-norm criterion. The three irreducible characters 1_G, sgn, χ_E are distinct, and S_3 has three classes, so completeness gives $\mathrm{Irr}(S_3) = \{\mathbf{1}, \mathrm{sgn}, E\}$.

Example 6.10 (Guard: norm one is not enough)**.** On S_3, $-\chi_E$ has norm $\langle \chi_E, \chi_E \rangle_{S_3} = 1$. By orthonormality,

$$\Big\langle \frac{1}{\sqrt{2}}(1_G + \mathrm{sgn}), \frac{1}{\sqrt{2}}(1_G + \mathrm{sgn}) \Big\rangle_{S_3} = \frac{1}{2}(1 + 0 + 0 + 1) = 1.$$

But a character takes the value $\dim V$, a non-negative integer, at e, and these take -2 and $\sqrt{2}$.

What does hold: let $\varphi = \sum_U a_U \chi_U$ with integers a_U and $\langle \varphi, \varphi \rangle_G = 1$. By orthonormality,

$$\sum_U a_U^2 = 1,$$

so one a_U is ± 1 and the others are 0: $\varphi = \pm \chi_U$. If also $\varphi(e) > 0$, then $\varphi = \chi_U$. Lecture 13 meets such an integral combination of norm one, and uses exactly this.

6.3 Tensor products, duals, Sym^2 and Λ^2

Definition 6.11 (Duals and tensor products)**.** Let (ρ, V) and (ρ', W) be representations of G.

- The *dual* is $V^* = \mathrm{Hom}(V, \mathbb{C})$, with $(g \cdot f)(v) = f(g^{-1} \cdot v)$. In the dual basis, g acts by the matrix $\rho^*(g) = D(g^{-1})^{\mathsf{T}}$.
- The *tensor product* is $V \otimes W$, with $g \cdot (v \otimes w) = (g \cdot v) \otimes (g \cdot w)$, so $(\rho \otimes \rho')(g) = \rho(g) \otimes \rho'(g)$.
- On $V \otimes V$, the *swap* is $\tau(v \otimes v') = v' \otimes v$. The *symmetric* and *exterior squares* are

$$\mathrm{Sym}^2(V) = \{x : \tau x = x\}, \qquad \Lambda^2(V) = \{x : \tau x = -x\}.$$

V^* is $\mathrm{Hom}(V, \mathbf{1})$ with Lecture 5's action, so it is a representation. $V \otimes W$ is one because $(A \otimes B)(A' \otimes B') = AA' \otimes BB'$.

Theorem 6.12 (Characters of constructions). *Let V and W be representations of G, and $d = \dim V$.*

(i) $\chi_{V^*} = \overline{\chi_V}$.

(ii) $\chi_{V\otimes W} = \chi_V \chi_W$.

(iii) $V \otimes V = \mathrm{Sym}^2(V) \oplus \Lambda^2(V)$, *a direct sum of subrepresentations of dimensions* $d(d+1)/2$ *and* $d(d-1)/2$, *and*

$$\chi_{\mathrm{Sym}^2(V)}(g) = \frac{1}{2}\big(\chi_V(g)^2 + \chi_V(g^2)\big),$$
$$\chi_{\Lambda^2(V)}(g) = \frac{1}{2}\big(\chi_V(g)^2 - \chi_V(g^2)\big).$$

Proof. (i) is Lemma 6.6 for $\mathrm{Hom}(V, \mathbf{1})$, as $\chi_{\mathbf{1}} = 1_G$.

Fix g. By Corollary 5.14(ii), V has a basis $v_1, \dots, v_d$ with $g \cdot v_i = c_i v_i$, and W a basis $w_1, \dots, w_{d'}$ with $g \cdot w_j = b_j w_j$.

(ii) g multiplies the basis vectors $v_i \otimes w_j$ of $V \otimes W$ by $c_i b_j$, so

$$\chi_{V\otimes W}(g) = \sum_{i,j} c_i b_j = \chi_V(g)\,\chi_W(g).$$

(iii) *The splitting.* τ is an intertwiner:

$$\tau\big(g \cdot v \otimes g \cdot v'\big) = g \cdot v' \otimes g \cdot v = g \cdot \tau(v \otimes v').$$

So $\mathrm{Sym}^2(V) = \ker(\tau - \mathbb{1})$ and $\Lambda^2(V) = \ker(\tau + \mathbb{1})$ are subrepresentations. As $\tau^2 = \mathbb{1}$, each x splits as

$$x = \frac{1}{2}(x + \tau x) + \frac{1}{2}(x - \tau x),$$

in $\mathrm{Sym}^2(V)$ and $\Lambda^2(V)$; only 0 satisfies $\tau x = x = -x$.

Bases. The vectors

$$v_i \otimes v_j + v_j \otimes v_i \quad (i \le j) \quad \text{in } \mathrm{Sym}^2(V),$$
$$v_i \otimes v_j - v_j \otimes v_i \quad (i < j) \quad \text{in } \Lambda^2(V)$$

span $V \otimes V$, and there are

$$\frac{1}{2}d(d+1) + \frac{1}{2}d(d-1) = d^2$$

of them, so they are bases of the two squares. g multiplies each by $c_i c_j$.

The characters. The eigenvalues of $\rho(g^2) = \rho(g)^2$ are the c_i^2, so $\chi_V(g^2) = \sum_i c_i^2$, and

$$\chi_V(g)^2 = \Big(\sum_i c_i\Big)^2 = \chi_V(g^2) + 2\sum_{i<j} c_i c_j.$$

So $\chi_{\Lambda^2(V)}(g) = \sum_{i<j} c_i c_j$ is the second formula, and the first is $\chi_{\mathrm{Sym}^2(V)} = \chi_V^2 - \chi_{\Lambda^2(V)}$, by (ii) and Proposition 6.2(iii). ∎

Example 6.13 (The squares of E). The squares of e, $(1\,2)$, $(1\,2\,3)$ are e, e, $(1\,3\,2)$, so by Example 6.4:

class of g	e	$(1\,2)$	$(1\,2\,3)$
$\chi_E(g)^2$	4	0	1
$\chi_E(g^2)$	2	2	-1
$\chi_{\mathrm{Sym}^2(E)}(g)$	3	1	0
$\chi_{\Lambda^2(E)}(g)$	1	-1	1

So $\mathrm{Sym}^2(E) \cong \mathbf{1} \oplus E$ and $\Lambda^2(E) \cong \mathrm{sgn}$, by Corollary 6.8(iii).

6.4 Lifting from quotients

Theorem 6.14 (Lifting). *Let $N \trianglelefteq G$, with quotient map $q\colon G \to G/N$.*

(i) If $\bar\rho$ is a representation of G/N on U, then $\rho = \bar\rho \circ q$ is a representation of G on U with $N \le \ker\rho$, and $\chi_\rho(g) = \chi_{\bar\rho}(gN)$. Every representation ρ of G with $N \le \ker\rho$ is $\bar\rho \circ q$ for exactly one $\bar\rho$.

(ii) $\bar\rho \circ q$ is irreducible if and only if $\bar\rho$ is, and $\bar\rho \circ q \cong \bar\rho' \circ q$ if and only if $\bar\rho \cong \bar\rho'$. So, up to isomorphism, lifting is a bijection from $\mathrm{Irr}(G/N)$ onto the irreducible representations of G with N in their kernel.

(iii) The linear characters of G are the lifts of the linear characters of G/G'.

Proof. (i) ρ is a composite of homomorphisms, trivial on $N = \ker q$, and $\rho(g) = \bar\rho(gN)$. Conversely, if $N \le \ker\rho$, then $\rho\colon G \to \mathrm{GL}(U)$ factors through G/N in exactly one way, by Theorem 2.3(i).

(ii) As q is onto, the operators $\rho(g)$ are exactly the operators $\bar\rho(gN)$. So the two have the same subrepresentations, and a linear map intertwines the lifts exactly when it intertwines $\bar\rho$ and $\bar\rho'$.

(iii) A linear character is a homomorphism $G \to \mathbb{C}^\times$, to an abelian group, so it factors through G/G', by Proposition 4.12(iii). Conversely, a lift of a linear character is linear, by (i). ∎

6.5 The size of character values

Proposition 6.15 (The size of character values). *Let (ρ, V) be a representation of G with character χ, and $g \in G$.*

(i) $|\chi(g)| \le \chi(e)$, with equality if and only if $\rho(g)$ is a scalar, $\rho(g) = c\mathbb{1}$.

(ii) $\chi(g) = \chi(e)$ if and only if $\rho(g) = \mathbb{1}$. So the kernel $\ker\chi = \{g : \chi(g) = \chi(e)\}$ *of χ is $\ker\rho$, a normal subgroup.*

Proof. If $V = 0$, both parts hold, as $\mathrm{End}(0) = \{0\}$ and $\chi = 0$; let $d = \chi(e) > 0$.

(i) By Corollary 5.14(ii), $\rho(g)$ is diagonalisable, with eigenvalues $c_1, \dots, c_d$ of modulus 1. By the triangle inequality,

$$|\chi(g)| = \left|\sum_i c_i\right| \le \sum_i |c_i| = d.$$

Equality holds exactly when the c_i are positive multiples of one another; being of modulus 1, they are then all equal to one c. A diagonalisable map with the single eigenvalue c is $c\mathbb{1}$. Conversely, $|\mathrm{tr}(c\mathbb{1})| = d$.

(ii) If $\chi(g) = d$, then $\rho(g) = c\mathbb{1}$ by (i), and $dc = d$ gives $c = 1$. The converse is clear, and $\ker\rho$ is normal, as the kernel of a homomorphism (§ 2.1). ∎

Example 6.16 (Equality outside the kernel)**.** In Lecture 5's irreducible plane for D_4, the central r^2 acts by $\rho(r^2) = -\mathbb{1}$, so

$$\chi(r^2) = -2 :$$

equality in (i), with $c = -1$, but $r^2 \notin \ker\chi$ by (ii).

Lecture 9's proof of Burnside's theorem rests on both parts.

6.6 Worked example

What is the character table of S_4?

The classes. Recall the cycle types in § 2.6. Five distinct irreducible characters will therefore complete the table (Corollary 6.8(v)).

The linear characters. $S_4' = A_4$ and $S_4/A_4 \cong C_2$ (§ 4.5), and C_2 has two linear characters (Example 5.15). By Theorem 6.14(iii), S_4 has two, 1_G and sgn.

The standard representation. On $\mathbb{C}^4$, $\chi(g) = |\mathrm{Fix}(g)|$ (Example 5.8), and

$$\langle 1_G, \chi_{\mathbb{C}^4}\rangle_{S_4} = \frac{1}{24}(4 + 6\cdot 2 + 3\cdot 0 + 8\cdot 1 + 6\cdot 0) = 1.$$

As for $\mathbb{C}^3$ (§ 5.5), $\mathbb{C}^4 = \mathbb{C}(1,1,1,1) \oplus W$, where

$$W = \{x : x_1 + x_2 + x_3 + x_4 = 0\}.$$

So $\chi_W = |\mathrm{Fix}| - 1$ (Proposition 6.2(iii)), and

$$\langle \chi_W, \chi_W\rangle_{S_4} = \frac{1}{24}(9 + 6\cdot 1 + 3\cdot 1 + 8\cdot 0 + 6\cdot 1) = 1.$$

By Corollary 6.8(iv), W is irreducible.

Its sign twist. The tensor-product character formula gives $\mathrm{sgn}\cdot\chi_W$. Its norm is unchanged, so $\mathrm{sgn}\otimes W$ is irreducible. The values on transpositions distinguish it from W.

The lift. § 2.6 gives a homomorphism $\pi : S_4 \to S_3$, onto, with kernel the Klein four-group V_4; by Theorem 6.14(ii), $\rho_E \circ \pi$ is irreducible. To read off its character, find the class of $\pi(\sigma)$:

- $\pi(\sigma) = e$ exactly for $\sigma \in V_4$, the class (2^2) with e;
- a transposition and a 3-cycle go to the same types,

$$\pi\big((1\,2)\big) = (2\,3), \qquad \pi\big((1\,2\,3)\big) = (1\,3\,2);$$

- $(1\,2\,3\,4) \notin V_4$, but

$$(1\,2\,3\,4)^2 = (1\,3)(2\,4) \in V_4,$$

so $\pi((1\,2\,3\,4))$ has order 2: a transposition.

So $\chi_E \circ \pi$ is read off Example 6.4.

The table. Five distinct irreducible characters complete the table.

cycle type	(1^4)	$(2,1^2)$	(2^2)	$(3,1)$	(4)
size	1	6	3	8	6
1_G	1	1	1	1	1
sgn	1	−1	1	1	−1
χ_W	3	1	−1	0	−1
$\operatorname{sgn}\cdot\chi_W$	3	−1	−1	0	1
$\chi_E \circ \pi$	2	0	2	−1	0

As a check, the rows are orthogonal:

$$\langle \chi_W, \chi_E \circ \pi \rangle_{S_4} = \frac{1}{24}(6 + 0 + 3 \cdot (-2) + 0 + 0) = 0.$$

The squares of the degrees add up to the order,

$$1 + 1 + 9 + 9 + 4 = 24 = |S_4|\,,$$

and Lecture 7 proves that $\sum_U (\dim U)^2 = |G|$ for every G.

Self-check. Find the kernel of each irreducible character of S_4. *Answer:* by Proposition 6.15(ii), read off where each row equals its degree:

χ	1_G	sgn	χ_W	$\operatorname{sgn}\cdot\chi_W$	$\chi_E \circ \pi$
$\ker\chi$	S_4	A_4	$\mathbf{1}$	$\mathbf{1}$	V_4

These recover § 2.6's normal subgroups.

Summary.

- *The main results:* $\dim \operatorname{Hom}_G(U,V) = \langle \chi_U, \chi_V \rangle_G$ (Theorem 6.7), so the irreducible characters are orthonormal and a representation is determined by its character (Corollary 6.8); the characters of constructions and lifts (Theorems 6.12 and 6.14); the values bound and the kernel (Proposition 6.15).
- *Handed on:* at most as many irreducibles as classes (Corollary 6.8(v)), to Lecture 7, which proves equality; the values bound and kernels to Lecture 9's Burnside theorem, and $\chi(g^2)$ to its indicator; the norm-one guard to Lecture 13; the table of S_4 to Lectures 8 and 13.
- *Next:* Lecture 7 asks where the irreducibles live, and why there are as many as classes.

References

Serre, *Linear Representations of Finite Groups*: characters, Props 1–3, pp. 10–11; the dual, Exercise 2.3, p. 12; orthonormality and multiplicities, Thms 3–4 and their corollaries, pp. 15–16; $|\chi(s)| \le n$, Exercise 6.7, p. 53. James and Liebeck, *Representations and Characters of Groups*: the values bound and $\ker\rho$, Thm 13.11, p. 124; $\dim \operatorname{Hom}_{\mathbb{C}G}(V,W)$, Thm 14.24, p. 145; lifting, Thm 17.3, p. 169, and linear characters, Thm 17.11, p. 174; the two squares, Prop. 19.14, p. 198; the table of S_4, §18.1, pp. 179–180.[2]

[2] James and Liebeck use right modules, $v \mapsto vg$, and write $\chi(1)$ for the degree.